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AI as Fifth Team Member: Differentiated Roles in Design Education

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Abstract

When individuals use generative AI, the diversity of ideas decreases sharply. When entire teams rely on the same AI tools, the convergence compounds. Most research to date has focused on enhancing individuals' creative abilities through use of generative AI. However, no prior studies have examined how distributing different AI tools across team roles affects creative collaboration. This research asks whether providing different AI capabilities to each role within a team can help counteract this trend toward homogenisation. This paper introduces the "AI as Fifth Team Member Model" for organising student design teams and their respective AI tools. The model defines four different roles: Research Coordinator (using ChatGPT or Claude), Visual Design Specialist (Midjourney or DALL-E), Space Planning Expert (Planner 5D or Homestyler), and Presentation Coordinator (who uses Canva AI or D5 Render). Each tool mediates creativity in a different way, through text-based analysis, image generation, 3D modelling and layout design, which forces heterogeneity into the team's creative process. Each role also produces a different kind of output, and combining them requires human deliberation. That is where the diversity comes from. A pilot of the model took place in two workshops at a Kristiania University of Applied Sciences: a four-day bachelor-level workshop and a two-day master-level workshop. The pilot results indicated that four days were not enough for bachelor-level design teams to form a cohesive unit, and that daily synchronisation protocols (i.e., morning sync, midday integration, and end-of-day synthesis) are essential for facilitating cross-role coordination. The model is currently being used in an educational context, but the way tasks are divided among roles is similar to how many professional design studios now divide tasks among specialists. Future studies should assess larger groups and measure formally the increased diversity in outputs, to verify these preliminary findings.

Keywords: human-AI collaboration, design education, generative AI, AI role differentiation, team creativity, higher education

Introduction

The vast majority of current design school approaches provide students with access to generative AI (GenAI), allow them to freely test and develop prompts independently, and ultimately evaluate student-created works alongside traditionally developed works. The academic literature reflects these methods: most studies focus on a singular student using a singular AI tool (Lu et al., 2025; Wagler, 2025); however, professional design practices inherently involve collaboration. Design firms, branding agencies and architectural firms typically operate with teams of specialists who separately focus on researching, creating visualisations, developing spatial plans, communicating with clients etc. Although there are significant differences between how students learn to use AI individually and in a professional setting, this discrepancy has not been addressed to date. Emerging evidence suggests that role differentiation occurs naturally in human-AI teams (Ju & Aral, 2025), yet how to structure this differentiation deliberately in educational settings remains unexplored. Recent empirical evidence highlights the urgency of this gap. Holzner et al. (2025) conducted a meta-analysis of 28 studies including N = 8,214 participants and concluded that while humans using generative AI performed significantly better than humans performing the task alone (Hedges' g = 0.27), the number of diverse ideas generated was significantly decreased (g = -0.86). As a result, when everyone relies on the same AI tool, their outputs tend to

converge. Xia et al. (2026) confirmed this at team level in that groups without GenAI support exhibited higher levels of group creativity - specifically in terms of relevance, elaboration, and originality - than groups provided with GenAI support in a factorial experiment with 176 university students. Luo et al. (2025) also found a similar double-edged pattern of homogenisation across 308 students in 66 groups: perceived AI intelligence boosts collective efficacy while simultaneously increasing task conflict. Given this data, it seems reasonable to conclude that if individual use of AI generates less diverse outputs, then we would anticipate that teams using the same AI tool would exacerbate this issue even further. Therefore, providing all students with "the same tool" is insufficient. Instead, we require models that distribute different types of AI capabilities among team members so as to maintain the creative benefits of each member while combating reduced variance in outputs. One such model is presented in this paper. The "AI as Fifth Team Member" framework distributes four unique AI-enhanced roles for each member of a four-person design team based upon their designated role: Research Coordinator, Visual Design Specialist, Space Planning Expert and Presentation Coordinator. This framework has been tested during two workshops at a Kristiania University of Applied Sciences. More specifically, this study examines whether different capabilities provided to each role within a team by using AI will provide a means to offset the trend towards structural similarity in solutions.

Related Work

The Diversity Paradox in AI-Augmented Creativity

There is a diversity paradox, that GenAI aids in increasing creativity yet decreases creative variety. Aru (2025) argues that there is a mechanism behind these phenomena. According to him, most generative models are trained on overlapping datasets and therefore rely on similar pattern pools. When two students use the same generative model but submit differing prompts, both may still produce responses that fall within the same solution area. In other words, while the lack of creative variety is not simply a result of a student being ineffective at developing effective prompts; it is inherent in the design of generative models.

In addition to individual level issues regarding the use of GenAI, the results reported by Xia et al. (2026) and Luo et al. (2025) demonstrate additional problems related to the use of GenAI. Specifically, Luo demonstrated that team trust strengthens the positive pathway (collective efficacy) but does not moderate the negative pathway (task conflict); therefore, even when teams express high levels of trust, they cannot eliminate the increased conflict created by the use of GenAI. These findings underscore the need for structural responses to the diversity paradox at the team level.

AI as Team Member: Theoretical Frameworks

Current theoretical frameworks describe possible ways in which AI can be used in educational settings in collaborative group work, however, there exists a lack of empirical testing of these models. Siemon (2022) used a multi-method approach (n = 1,358 survey respondents and nine expert interviews) to construct four team roles that AI-based teammates can fulfil: Coordinator, Creator, Perfectionist, and Doer. More recently, Tummala et al. (2025), drawing on Shared Mental Models (Cannon-Bowers & Salas, 2001) and Co-Active Design (Johnson et al., 2014), proposed a broader conceptual framework with three categories of AI participation in groups: Supporting Individual Human Team Members, Supporting the Team as a Whole, and performing as an Equivalent Team Member. However, neither Siemon's four roles nor Tummala et al.'s three categories have been tested empirically in creative education settings.

Most previous research has focused on decision-making processes in teams involving AI rather than on the creative process. Centralising AI knowledge has been shown to improve decision quality (Zercher et al., 2025). Decentralising AI knowledge, without a protocol for integrating the knowledge into a cohesive unit, can lead to a decrease in team-trust, biased information processing, and a shift away from cooperative problem-solving to a more critical approach (Zercher et al., 2025). No study to date has tested distributing different AI specialisations across human team members in a creative context. Chen et al. (2025) came closest: in a within-group experiment with 15 design students, a team of three specialised AI agents (Creator, Expert, and Doer) outperformed a single general-purpose agent on trust, satisfaction, and team performance. However, their specialisation was distributed across AI agents, not across human team members - the configuration proposed in this paper.

A five-level agency spectrum (Passive, Reactive, Semi-Active, Proactive, Cooperative) for Human-AI Co-Creation was described by Zhang et al. (2025) in a scoping review of 134 studies. Zhang et al. noted that most current research is concentrated at the reactive and passive levels of the spectrum, leaving semi-active and proactive forms of agency comparatively underexplored. As part of a qualitative analysis of a single AI-in-use case study,

Grange et al. (2026) observed that, during a design sprint, AI systems tended to assume convergent tasks, while humans retained divergent thinking. However, the division of labour between humans and AI in terms of task assignment remains only weakly theorised.

The Missing Piece: Pedagogical Models

Although the frameworks discussed above have been developed, there are currently no studies published that test how students can be differentiated by structured AI roles in a creative educational environment. The literature on design education using AI has remained within an individual perspective - as with each student, each tool (Lu et al., 2025; Wagler, 2025). Research on design workshops has examined how duration and group formation affect student experience (Primus & Sonnenburg, 2024); however, there is no consideration of structuring AI use as part of collaborative arrangements. Gibson et al. (2023) have suggested a three-tiered model of AI in learning - micro (student), meso (team), macro (cultural/institutional) - and although they have provided some examples of what could occur at each tier, the team (meso) tier has largely been left without practical examples or applications. The model presented in this paper operates at this meso level.

The "AI as Fifth Team Member" Model

Design Rationale

A structural response is required to resolve the diversity paradox at the system level. Meincke et al. (2024) demonstrated that individual Chain-of-Thought prompting increases the diversity of AI-generated ideas. However, this has no bearing on the issue of convergence when all members of a team rely on the same Chain-of-Thought prompt engine. Therefore, if convergence stems from reliance on the same source (in this case, an AI system), then a logical response is to reduce that shared reliance.

There are many examples outside of AI where creative tools define their own logics of form. For example, when architects were first introduced to NURBS modelling software (that was initially designed for automotive applications and film VFX), it defined the logic of animation and therefore shifted architectural thinking from being based upon a rectilinear geometry of modernism to being defined by smooth, curvilinear forms. ZBrush was initially a painting application with added depth and evolved into a digital sculpting tool, creating a completely new artistic domain (Hanzl, 2019). The same principle extends to generative AI. One tool reads a design brief and works with it in text; another interprets it visually, generating images; a third takes the same brief into three dimensions, laying it out as floor plans and volumes. In other words, each tool defines its own logic of creation, and each generates output that cannot be combined using another AI. Human deliberation is required to combine these tools - the team must determine what works, what conflicts and what to maintain. It is here in the deliberation that we preserve diversity.

Role Structure

This model assigns each of the four members of a team one of four roles. Each role is paired with AI tools suited to its function (Figure 1): text-based analysis for research, image generation for visualisation, 3D planning for spatial layout, and presentation tools for client-facing output. While outputs flow sequentially from research, through visualisation and spatial planning, and then into the presentation phase, there are opportunities for feedback as well. The Research Coordinator may refine the design brief based on what the Visual Designer produces using Midjourney. The Space Planner may adjust the layout based on the visual concepts developed by the Visual Designer. The "Fifth Team Member" analogy illustrates another aspect of how teams using AI have to function. In a team consisting solely of humans, integration occurs through natural interaction. When AI generates part of the output, that integration must be deliberately structured and coordinated, evaluated, and combined just like any other teammate's contribution. No individual has all the information. The full picture can only be achieved when everyone shares their pieces.

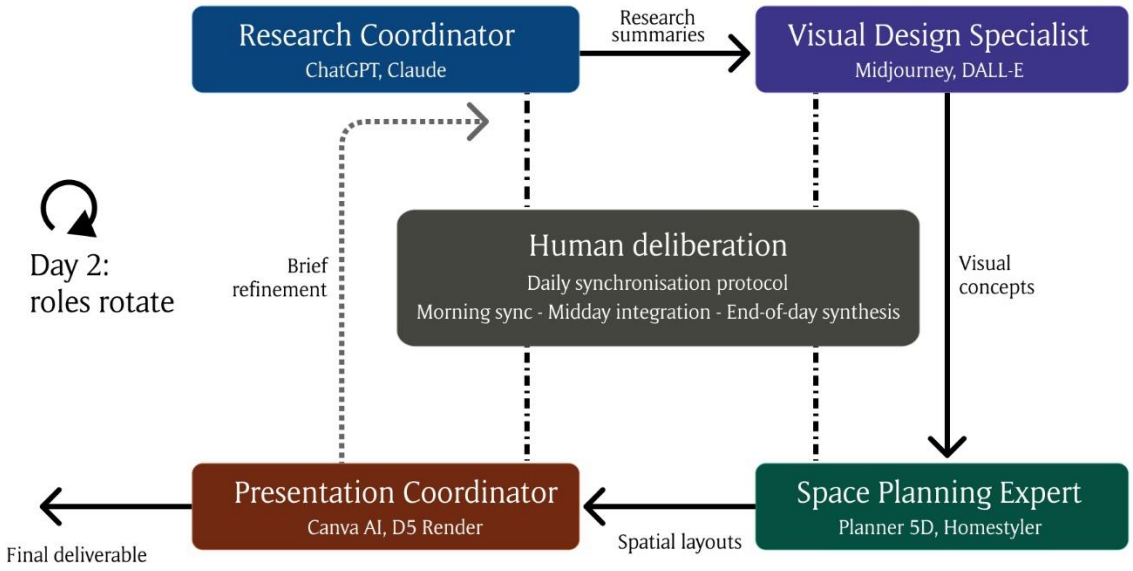


Figure 1: The "AI as Fifth Team Member" model - four roles, sequential output flow, and daily synchronisation protocol. Source: Authors' own work.

Synchronisation Protocol

Cross-role coordination follows a daily three-part structure. In a morning synchronisation meeting (15 minutes), all roles report what output has been generated by their artificial intelligence tools and what the team needs to prioritise. In a midday integration meeting (30 minutes), synthesis primarily occurs. All outputs are laid side-by-side and each owner signs off that they are passing on information to the next role. For example, in order for a visual designer to take action based on research findings, the researcher must first confirm whether the research findings are accurate (i.e., not simply hallucinations). At the end of the day (synthesis meeting - 15 minutes), the team documents what was completed and identifies any areas that will need to be addressed at the next morning meeting. This protocol is how distributed AI contributions become shared team knowledge. Asymmetrically distributed AI knowledge, if left unstructured, creates mistrust instead of fostering teamwork as demonstrated by Zercher et al. (2025). Similarly, Feng (2025) found that when students used AI chatbots for academic purposes, they interacted with the chatbot more than with their fellow classmates. Our use of a synchronised format encourages peer-to-peer interaction within the work process.

Role Rotation

On the second day, students switch to another role. In addition to fairness, this allows each student to have hands-on experience with at least two AI tools. No one will be restricted to only one type of role. The rotation of roles also creates a form of comparative literacy: a student that has researched using ChatGPT and created visuals using Midjourney can describe in detail how the use of each AI tool changes the way they create. That ability to compare how these tools shape their creativity is itself a learning outcome.

Pilot Implementation

The model was piloted on a small scale in two workshops at a Kristiania University of Applied Sciences in September 2025. No pre-post measures were used; the observations below are exploratory.

Hanzl developed and moderated the AI workshops, which were integrated into two previously established design courses run by Johansen.

The first workshop was held over four days (non-consecutive), 15-19 September 2025 (six hours daily across four days, 24 hours total). It was part of a bachelor-level course in Exhibition and Retail Design where students

develop independent practice-based design projects. There were three teams created from eleven students (including three international students).

Students had to create a sustainable co-working space redesign introducing more than 15 different AI tools to their projects. Roles rotated on Day 2 so that each student experienced more than one role.

The second workshop was held over two consecutive days (25-26 September 2025, 12 hours), as part of a master-level course in Visual Communication Design, which analyses how technologies influence the visual communication process - from craft-based techniques to advanced digital tools. Three pairs of students participated along with a “floating” observer-documenter. Two shorter briefs were used. Each pair participated in a High-Tech/Low-Tech partner structure, with role switching occurring at midday.

Preliminary Observations and Discussion

There were three primary trends identified in the data from each workshop. First, bachelor-level teams required more than four days to develop the same level of collaborative working relationships as master-level students, who established effective partnerships by day two. Gersick’s (1988) punctuated equilibrium theory proposes that teams undergo a major transition at the temporal midpoint of a project. However, for bachelor-level students the first two days were largely devoted to learning how to use AI tools and therefore had limited opportunity to work collaboratively prior to this transition.

Secondly, in the absence of a communication protocol, students in general worked independently with their respective AI tools. As noted within Section 3.3, the synchronisation protocol was critical for establishing collaborative work.

Thirdly, participants shifted from seeing AI as a form of cheating to viewing it as an aid. This shift appears primarily cognitive in nature: participants accepted the idea of using AI in principle, however some anxiety remained - consistent with Sat’s (2025) findings that attitudes improve but anxiety does not decrease. The model’s role structure reflects how professional design studios, game development teams, and advertising agencies distribute AI capabilities, suggesting potential transferability beyond education.

However, both samples are small (n = 11, n = 7) and drawn from a single institution, with one author serving as both designer and observer. These findings should be treated as preliminary.

Conclusion

An alternative to the “one-size-fits-all” paradigm is our proposed conceptual framework of “AI as Fifth Team Member”. The main idea behind this framework is to assign an appropriate AI capability (or set of capabilities) to match the skills/role of each person within the creative team. The success of our early experiments using this framework appears to be highly dependent upon the need for clearly defined procedures for coordinating how each individual uses their assigned AI tools when working collaboratively. The individuals’ roles need to rotate so they can experience both the benefits of receiving the assistance from the AIs and also the challenges associated with relying on themselves.

Competing interests

The authors have no competing interests to declare.

Ethics and consent

The workshops were conducted as part of regular course activities at Kristiania University of Applied Sciences. Participation in the AI-enhanced components was integrated into existing course curricula. All observations reported are exploratory and based on the workshop facilitator's field notes. No personally identifiable student data were collected or reported.

Authors' contributions

RH designed and facilitated both AI workshops, developed the "AI as Fifth Team Member" model, and drafted the manuscript. JJ coordinated the integration of the workshops into the existing course curricula, contributed to the conceptual development of the model, and reviewed the manuscript.

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